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RC-110 PRE-REGISTRATION DOCUMENT Floquet Logical Dynamics: Automorphism-Driven Waves on the [[24,12,8]] Quantum Golay Code

Date: 2026-07-07 Framework: 24D-DMF v8.4.2 Research Cycle: RC-110 Status: PRE-REGISTERED — Awaiting Execution

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1. MOTIVATION AND CONTEXT ==

1.1 Background

The 24D-DMF framework has established a mathematical skeleton consisting of:

- The extended binary Golay code C24 (classical and quantum [[24,12,8]])
- The Leech lattice Λ24 via Construction A
- The non-abelian engine G = C23   C11 (order 253)
- The S-involution S: x → -1/x (mod 23) as a confirmed M24 automorphism
- RC-100: S acts as a non-trivial logical Clifford gate on the 12 logical qubits
- RC-77: Logical Hadamard   P23 generates dihedral group D23 (order 46)
- RC-108: Stabilizer-flow projective dynamics converges to logical subspace

1.2 The Pivot

Previous cycles (RC-45 through RC-109) attempted to derive physical quantities (nucleon mass ratio 727/726, photon emergence, gauge zero modes) from static mathematical structure. This approach encountered a fundamental obstruction: static objects cannot produce dynamics without a dynamical principle being imposed.

RC-110 pivots to a different question: **What dynamics does the mathematical skeleton generate when treated as a quantum information system?** Specifically:

- The Golay code is a quantum error-correcting code (crystal)
- The Leech lattice is a sphere-packing lattice (larger crystal)
- The automorphism group M24 contains mechanical operations (permutations)
- The logical space is the “internal” gauge/charge degree of freedom
- The question: does mechanical deformation (automorphism action) couple to logical state evolution (piezoelectric-like effect)?

1.3 Physical Metaphor

Crystal: [[24,12,8]] stabilizer code with local constraints Phonon: low-energy excitation = syndrome violation = anyon Piezoelectric: mechanical stress (automorphism) → polarization (logical rotation)

This metaphor is INTERPRETIVE. The mathematical test is: does the engine automorphism group, acting on the physical qubits, induce non-trivial dynamics on the logical subspace?

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2. PRE-REGISTERED HYPOTHESIS

H1 (Primary): The composed Floquet operator U = H_L   P23   P11^(t), where H_L is the logical Hadamard and P23, P11 are engine automorphisms implemented as physical permutations, generates non-trivial logical state evolution with period 46 = 2×23 on the logical space.

H2 (Secondary): The full group generated by P23, P11, S on the logical space is larger than D23 and may generate the full logical Clifford group (or a significant subgroup thereof) on 12 qubits.

H3 (Tertiary): If H1 holds, the logical state trajectory over one full period (46 ticks) forms a discrete "wave" on the 4096-dimensional logical space, with interference patterns that depend on the engine schedule.

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3. CONSTRUCTION DEFINITIONS

3.1 The Quantum Golay Code

Stabilizer group S = {i^(wt(c)/2) X^c Z^c : c   C24}

- 12 stabilizer generators (from rows of generator matrix G24)
- Logical space: N(S)/S, dimension 2^12 = 4096
- Normalizer N(S) = {(a,b) : a+b   C24}

3.2 Physical Automorphisms

P23: cyclic shift (i → i+1 mod 23), fixes position 23 P11: multiplicative automorphism (i → 2i mod 23), fixes position 23 S: involution (x → -1/x mod 23 for x {1,...,22}, S(0)=23, S(23)=0)

All three are confirmed elements of M24, the automorphism group of C24.

3.3 Logical Operator Representation

Each automorphism g   M24 acts on Pauli operators by conjugation: g: X^a Z^b → X^(Ag + Cg) Z^(Bg + Dg) where the symplectic matrix M_g = [[A,B],[C,D]]   Sp(24, F2) characterizes the logical action.

For the [[24,12,8]] code, this induces an action on the 12 logical qubits: M_g^logical = [[A_L, B_L], [C_L, D_L]]   Sp(12, F2)

3.4 The Floquet Operator (One Tick)

At tick t, the Floquet operator is: U_t = H_L ·U_{g_t} where:

- g_t = P23^a ·P11^b with a = t mod 23, b = t mod 11
- U_{g_t} is the physical unitary implementing g_t

- H_L is the logical Hadamard (order 2, swaps X̄ Z̄)

The full period is LCM(23, 11, 2) = 506 ticks (if H_L is applied every tick).

Alternative schedules to test: (a) H_L applied every tick: period 506 (b) H_L applied every 23rd tick: period 253 (c) H_L applied every 11th tick: period 46

3.5 The Logical State Trajectory

Starting from a logical basis state | _0 in the 4096-dimensional space: | _{t+1} = U_t | _t

After T ticks, the state is: | _T = U_{T-1} ·U_{T-2} ·... ·U_0 | _0

The logical state trajectory is the sequence {| _t } for t = 0, 1, 2, ...

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4. FALSIFICATION CRITERIA

F1.1: Logical Periodicity Criterion: The logical state returns to itself (up to global phase) after some finite number of ticks. Threshold: | _0| _T |² > 0.99 for some T   1000. If FAIL: The dynamics is not periodic on the logical space.

F1.2: Non-Trivial Logical Evolution Criterion: The logical state changes non-trivially during the period. Threshold: max_t | _0| _t |² < 0.9 for some t < T. If FAIL: The automorphisms act trivially on the logical space (despite RC-100 showing S is non-trivial, the composed dynamics might collapse to identity).

F1.3: Engine Schedule Dependence Criterion: The logical trajectory depends on the engine schedule (P23, P11 ordering). Threshold: Two different schedules produce different logical periods or different trajectories from the same initial state. If FAIL: The logical dynamics is insensitive to the engine structure.

F1.4: State Space Coverage Criterion: The trajectory explores a non-trivial subset of the 4096-dimensional logical space. Threshold: The trajectory visits > 2 distinct logical basis states (or has non-zero amplitudes on > 2 basis states at some tick). If FAIL: The dynamics is confined to a subspace of dimension   2.

F1.5: Interference Structure Criterion: The logical state exhibits interference (superposition) at intermediate ticks, not just basis-state hopping. Threshold: At some tick t, the state has   3 non-zero amplitudes with distinct phases (not all real and positive). If FAIL: The dynamics is classical permutation of basis states.

F1.6: Comparison to Trivial Dynamics Criterion: The engine-driven dynamics differs from random logical operations or trivial dynamics. Control: Compare against (a) random Clifford gates, (b) identity evolution, (c) P23 alone (no P11, no H_L). Threshold: The engine-driven trajectory has a distinct period structure not seen in controls.

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5. METHODOLOGY

5.1 Step 1: Build the Quantum Golay Code

- Construct generator matrix G24 (cyclic or QR basis)
- Verify: self-dual, minimum weight 8, weight distribution (1, 759, 2576, 759, 1)
- Build stabilizer generators S_i = X^{c_i} Z^{c_i} for each row c_i of G24

- Build logical operators X̄_j, Z̄_j from cocode representatives

5.2 Step 2: Compute Symplectic Action

- For each g {P23, P11, S}, compute the 24×24 symplectic matrix M_g
- Restrict to logical space: extract 12×12 blocks A_L, B_L, C_L, D_L
- Verify: M_g^logical Sp(12, F2)
- Verify RC-100 result: S has B_L 0 (non-trivial logical action)

5.3 Step 3: Implement Floquet Dynamics

- Choose schedule: g_t = P23^(t mod 23) ·P11^(t mod 11)
- Apply H_L every k-th tick (k {1, 23, 11, 46} to test)
- Track logical state | _t as a 4096-dimensional complex vector
- Use sparse representation (stabilizer state evolution via Gottesman-Knill)

5.4 Step 4: Analyze Trajectory

- Compute return probability | _0| _t |^2 for t = 0 to T_max
- Identify period T where return probability > 0.99
- Compute state entropy S(t) = -tr(_t log _t) where _t = | _t   _t|
- Compute participation ratio PR(t) = 1/Σ| i| _t | (effective dimension)
- Fourier analyze return probability to identify frequency components

5.5 Step 5: Control Comparisons

- Control A: Random Clifford sequence (same length, random gates)
- Control B: P23 only (no P11, no H_L)
- Control C: P11 only (no P23, no H_L)
- Control D: H_L only (no automorphisms)

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6. EXPECTED OUTCOMES AND INTERPRETATION

6.1 If H1 is CONFIRMED (period-46 logical dynamics)

The engine's automorphism group, combined with logical Hadamard, generates periodic logical state evolution. This is a genuine dynamical mechanism built from the code's own structure — no fitted parameters, no target numbers. The "piezoelectric" metaphor is mathematically realized: physical permutations induce logical rotations.

Interpretation: The Golay code has internal dynamics when driven by its own automorphism group. The Leech lattice (via Construction A) may inherit similar dynamics at the lattice level.

6.2 If H2 is CONFIRMED (large logical group)

The group P23, P11, S generates a significant subgroup of the logical Clifford group. This would mean the code's own symmetry group is a computational resource — the automorphisms are not just passive permutations but active logical processors.

6.3 If F1.1 FAILS (no periodicity)

The logical state never returns to itself. The dynamics is either quasi-periodic or chaotic on the logical space. This would be surprising given the finite group structure, but possible if the Floquet operator has irrational eigenphases.

6.4 If F1.2 FAILS (trivial evolution)

Despite individual automorphisms being non-trivial (RC-100), their composition in the Floquet sequence collapses to identity. This would mean the "piezoelectric" effect cancels out over time — the code is dynamically inert.

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7. DELIBERATE EXCLUSIONS

To prevent the "target-seeking" bias identified in previous cycles:

- NO comparison to 727/726, proton/neutron masses, or any particle data
- NO attempt to derive , , or any fitted parameter from the dynamics
- NO pre-registered "success" criterion that requires matching a known physical constant
- NO use of the Gram matrix eigenvalues as energy levels (they are structural, not dynamical)
- NO claim that this dynamics "is" gravity, electromagnetism, or any known force

The only claim tested: **the mathematical skeleton generates dynamics when treated as a quantum information system.**

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8. POST-EXECUTION ANALYSIS

After execution, the following questions will be addressed:

Q1: What is the actual period of the logical state evolution? Q2: Does the trajectory depend on initial logical state? Q3: What is the structure of the logical operator generated over one full period? Q4: Does S (the involution) change the dynamics qualitatively when included vs. excluded? Q5: What happens if we extend to the full [[48,12,?]] coupled CSS code (RC-106) — does the dynamics double in complexity or collapse?

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9. CONNECTION TO FRAMEWORK HISTORY

RC-110 builds directly on:

- RC-77: D23 clock from logical Hadamard   P23 (order 46)
- RC-81/82: Non-abelian engine G = C23   C11 (order 253)
- RC-96: S-involution as M24 automorphism
- RC-100: S as non-trivial logical Clifford gate
- RC-108: Stabilizer-flow projective dynamics

RC-110 differs from:

- RC-45–RC-69: No attempt to match 727/726 or any particle mass
- RC-72–RC-76: No syndrome-feedback or Floquet-with-CNOT construction
- RC-83–RC-90: No perturbation theory or eigenvalue fitting
- RC-107–RC-109: No Hamiltonian construction on logical space

RC-110 is a pure dynamics test: **what does the skeleton do when you wind it up and let it go?**

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10. HONEST CLOSURE PREVIEW

Possible verdicts:

PASS (Minimum): F1.1 and F1.2 both pass — periodic, non-trivial logical dynamics confirmed.

PASS (Strong): F1.1–F1.4 pass — the dynamics explores the logical space non-trivially and is engine-dependent.

PASS (Full): F1.1–F1.6 pass — interference, non-classical dynamics with structure distinct from random controls.

FAIL (Structural): F1.1 or F1.2 fails — the code is dynamically inert under automorphism driving.

FAIL (Methodological): Computationally obstructed (4096×4096 operations too large) — requires sparse/efficient methods.

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END OF PRE-REGISTRATION

Pre-registered by: [Researcher] Date: 2026-07-07 Framework: 24D-DMF v8.4.2 Research Cycle: RC-110

This document is binding. No post-hoc criteria may be added. No target numbers may be introduced during execution. The analysis must report results exactly as defined above.

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